RAMYA K

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PROFESSIONAL SUMMARY

Java Backend Developer with 4+ years of experience building and supporting enterprise applications using Java, Spring Boot, microservices, Kafka, Hibernate, and SQL. Experienced in developing secure RESTful APIs, implementing OAuth 2.0 and JWT authentication, integrating distributed systems, and deploying containerized applications to cloud environments. Strong background in performance optimization, automated testing, production monitoring, and incident resolution.

TECHNICAL SKILLS

- **Programming Languages**: Java 8, SQL, JavaScript (ES6+), TypeScript, C#, Python, Kotlin
- **Java & Backend**: Spring Boot, Spring Security, Spring WebFlux, Spring AOP, Spring Security, Hibernate, Rest APIs, Microservices, .NET Core, FastAPI, Node.js, GraphQL, Resilience4J, JPA
- **Frontend**: Angular, React.js, HTML5, CSS3, NgRx, Svelte, Reactive Forms, Angular Material, RxJS
- **Databases**: MySQL, PostgreSQL, MongoDB, DynamoDB
- **Cloud and DevOps**: AWS, Azure, Docker, Kubernetes, GitHub Actions, Azure DevOps
- **Testing**: JUnit, Mockito, Postman, Playwright, SOAPUI, Swagger/Open API
- **Monitoring & Tools**: Git, Maven, Splunk, Zipkin, SonarQube, ELK Stack
- **AI & LLM Technologies**: Azure AI Document Intelligence, Azure AI Language, Vertex AI, LangChain, Retrieval-Augmented Generation (RAG), Hugging Face Transformers, Pinecone, FAISS, TensorFlow, Keras, Vector Search, LangFlow

PROFESSIONAL EXPERIENCE

State of Ohio | ***Software Developer*** **May 2026 - Present**

- Collaborate with business stakeholders, product owners, analysts, and solution architects to analyze public-sector workflows, define technical requirements, identify system integrations, and translate functional requirements into detailed technical designs that align system solutions with business needs
- Develop enterprise application features using object-oriented programming, layered architecture, dependency injection, reusable service components, and automated unit testing, applying backend engineering principles to improve application maintainability and reliability
- Build and integrate RESTful services, leveraging Swagger for API documentation, unit tests for validation, and robust exception handling, which enabled reliable data exchange and reduced integration errors.
- Develop responsive front-end applications using Angular, TypeScript, HTML, CSS, and NgRx, following component-based design and WCAG accessibility standards to improve usability and support accessible user experiences
- Implement secure application workflows using JWT authentication, claims-based authorization, role-based access control, and protected API communication to safeguard application data and enforce secure access
- Develop an AI-assisted document-processing microservice using Python, FastAPI, Azure AI Document Intelligence, and Azure AI Language to extract content and redact personally identifiable information, incorporating confidence-based human review and role-based access controls, which speeds up document handling and reduces manual redaction effort

NYSIF | ***Software Engineer*** **Feb 2025 - May 2026**

- Developed backend and server-side applications for policy administration and claims processing with Java 11, Spring Boot, and Kotlin components, using VS Code and PostgreSQL; the services handled statewide insurance workflows and high-volume daily transactions, ensuring reliable processing for all applicable users.
- Designed and implemented RESTful microservices with clearly defined service boundaries and API contracts, enabling independent development, testing, and deployment across application components.
- Built asynchronous, event-driven integrations using GCP Pub/Sub to process policy and claims events, reduce direct dependencies between services, and improve application resiliency under high transaction volumes.
- Implemented secure access to internal and partner-facing APIs using Spring Security, OAuth 2.0, JWT, Azure AD, and role-based authorization, following organizational data-governance requirements for sensitive policy and claims information.
- Containerized and deployed backend services using Docker, Cloud Run, and Google Kubernetes Engine, integrating Cloud SQL and Cloud Storage to support reliable application data storage and centralized document management
- Established CI/CD workflows using GitHub Actions and GCP deployment tools to automate builds, testing, and application deployments across environments.
- Monitored production services using GCP Cloud Logging and Cloud Monitoring, investigating application errors, latency issues, and service disruptions to support reliable system operation.
- Designed and operationalized ML-backed services using Vertex AI to support claims risk scoring and predictive analytics, shortening underwriting decision time and improving consistency in risk evaluation.

Capital One | ***Full Stack Software Engineer*** **Jun 2024 - Jan 2025**

- Designed and developed cloud-native microservices using Java & Spring Boot to support digital banking and payment workflows for 400K+ retail users, ensuring secure transactions, high availability, and consistent system uptime.
- Built reactive services using Spring WebFlux and implemented circuit breakers, retries, and timeouts with Resilience4J to improve throughput and prevent cascading failures during downstream service disruptions.
- Implemented asynchronous, event-driven data infrastructure using Kafka to process events asynchronously, reducing interservice data exchange latency and improving system reliability under peak traffic.
- Optimized REST and GraphQL APIs through SQL query tuning, indexing, and data-access improvements, reducing average response times by approximately 25% across key customer-facing endpoints.
- Implemented OAuth 2.0, JWT authentication, and role-based authorization to secure customer and transaction data across internal and customer-facing APIs.
- Containerized and deployed services using Docker and Kubernetes on AWS, supporting scalable releases, service monitoring, and recovery during production incidents.
- Participated in on-call rotations, diagnosing API timeouts, Kafka consumer lag, and downstream service failures through application logs, metrics, and distributed traces to restore production services.

Tata Consultancy Services | ***Software Developer*** **May 2021 - Jul 2022**

- Modernized a large-scale telecom subscription and billing platform supporting 600K+ accounts by refactoring legacy monolithic components into a microservices-based architecture, improving scalability and long-term maintainability.
- Built and optimized backend services using Java and Spring, applying core concepts including multithreading, collections, exception handling, and functional programming to improve reliability and performance of billing workflows.
- Designed and implemented data access layers using Hibernate and Spring, leveraging HQL, Criteria queries, batch processing, and second-level caching to reduce database load and improve response times by ~30%.
- Developed RESTful APIs with Spring Boot while maintaining SOAP integrations (JAX-WS) for interoperability with legacy systems, ensuring seamless migration without service disruption.
- Applied Java concepts including multithreading, collections, exception handling, and functional programming to improve the reliability and performance of high-volume transaction processing.
- Implemented structured logging and lightweight health and metrics endpoints, reducing incident triage time by ~25% during high-volume billing and invoicing cycles.
- Built data pipelines to process millions of records daily using Python and FastAPI, optimizing batch processing and reducing manual data handling effort by ~60%.

Vitech Systems Group | ***Junior Software Engineer*** **Jan 2020 - Mar 2021**

- Developed and maintained backend services using Java, Spring Boot, and Spring MVC for a policy administration platform supporting more than 50K policyholders.
- Built RESTful APIs and integrated them with React-based dashboards to support core policy-management workflows.
- Developed database components using PostgreSQL, including SQL queries, schema updates, data validation, and application-level persistence operations, supporting reliable policy data management
- Investigated production issues by analyzing ELK Stack logs, debugging Java services, and validating database records to restore stable application behavior.

PROJECTS

AI-Powered Personal Knowledge Engine **Jan 2026**

- Built a RAG-based knowledge system to ingest documents, source code, and notes and provide context-grounded answers using vector search and metadata-aware retrieval.
- Developed modular ingestion, chunking, embedding, retrieval, and response-generation services, with source attribution and configurable retrieval strategies.
- Evaluated chunking, ranking, retrieval relevance, latency, and hallucination behavior to improve answer quality and system observability.

Adaptive Industrial Anomaly Detection and Localization

- Built an unsupervised visual inspection pipeline that learned normal product representations from defect-free images to detect and pixel-localize previously unseen manufacturing defects.
- Implemented and compared convolutional autoencoder, PatchCore, and self-supervised Vision Transformer models across multiple industrial object and texture categories.
- Targeted at least 97% image-level AUROC, 92% pixel-level AUPRO, and a 20% reduction in false alarms at matched recall through test-time adaptation and calibrated uncertainty.

EDUCATION

State University of New York at NEW PALTZ | ***Master of Science*** **New York, USA May 2024**

RVR&JC College Of Engineering | ***Bachelors of Technology*** **Andhra Pradesh, India Jun 2021**